

Predicting the Founding of Social Organizations Among ~~Tecnológico de Monterrey~~ Alumni: A Machine Learning Approach

Gerardo Leiva Díaz (A01198479)
Isaac Francisco Sánchez Veloquio (A01198327)
David Alejandro Lozano Arreola (A01722728)
Jorge Betanzo Carriles (A01198676)¹

¹Tecnológico de Monterrey

June 2026

Abstract

✓ This paper develops a machine learning pipeline to identify which demographic and academic characteristics of Tecnológico de Monterrey alumni are most associated with founding non-profit social organizations. Using a survey dataset of $N = 25,356$ alumni collected on the institution's 80th anniversary in collaboration with the ranking agency Quacquarelli Symonds (QS), we applied preprocessing, exploratory data analysis, and six classification models. Gradient Boosting achieved the highest discriminative performance (ROC-AUC = 0.801, 5-fold CV). SHAP analysis reveals that volunteering behavior, prior company founding, age, philanthropic donation habits, and leadership experience are the dominant predictors. Alumni who score highly on entrepreneurship competency self-assessments and report a strong sense of purpose in life are also substantially more likely to participate in the foundation of social organizations. These findings support the formulation of early identification programs for current students.

Keywords: social entrepreneurship; machine learning; gradient boosting; SHAP; alumni impact; non-profit founding.

1. Introduction

Tecnológico de Monterrey has built an 80-year tradition of academic excellence alongside a commitment to social impact. As part of its anniversary celebrations, a large-scale longitudinal study conducted by QS surveyed the global EXATEC community to assess how alumni contribute to society beyond their professional career [1]. One of the most visible forms of social contribution is the founding of non-profit or civil-society organizations.

Understanding which alumni profiles predict this behavior is valuable for two reasons. First, it allows the institution to identify and support students who are more likely to

become social entrepreneurs. Second, it provides empirical evidence for the hypothesis that specific academic and demographic trajectories foster civic engagement.

We frame this as a binary classification problem: **given a set of demographic, academic, career, and wellbeing features, can we predict whether an alumnus has founded a non-profit organization?** The outcome variable, `founded_nonprofit`, is highly imbalanced (6.95% positive), which we handle through class-weight balancing and evaluation metrics suited to imbalanced data (ROC-AUC, F1).

The remainder of the paper is organized as follows. Section 2 describes the dataset and preprocessing pipeline. Section 3 presents the modelling methodology. Section 4 reports results and model comparison. Section 5 details the SHAP interpretation. Section 6 characterises the high-probability alumni profile. Section 7 discusses findings and limitations, and Section 8 concludes.

2. Data and Preprocessing

2.1. Dataset

The primary dataset is the *Base de respuestas 80 aniversario QS completa* survey, containing 25,356 complete responses across 223 raw variables. A complementary data dictionary and a longitudinal 75th-anniversary dataset were used as reference. The survey was administered via Qualtrics in early 2023 and covers graduate alumni from all Tec campuses from 1952 to 2022.

2.2. Target Variable

The binary target is the response to: “Since graduating from Tecnológico de Monterrey, have you founded a non-profit organization, as part of the founding group or main founder?”

Table 1: Target variable class distribution.

Class	Count	%
Did not found	23,593	93.05
Founded nonprofit	1,763	6.95
Total	25,356	100.00

2.3. Feature Engineering

Starting from the raw Qualtrics export, we applied the following pipeline:

1. **Column selection:** 64 semantically significant features that covered demographics, academic background, career trajectory, leadership history, social engagement, competency self-assessments, and wellbeing scales.
2. **Missing value handling:** Columns with more than 60% missing were dropped (`years_abroad`, `volunteer_hours_month`, `donation_amount`). The remaining numeric gaps were imputed with the column median; categorical gaps with the column mode.
3. **Binary encoding:** Survey responses to Yes/No mapped to 1/0. Ordinal levels of parent-education encoded numerically.
4. **Engineered features:** `years_since_grad` (survey year minus graduation year), aggregate competency scores, competency growth from graduation to present, leadership breadth (count of senior roles held), and salary growth ratio.
5. **One-hot encoding:** 14 nominal categorical variables (nationality, program, school, employment status, etc.) expanded to a final feature matrix of 544 columns.

Table 2 summarizes the descriptive statistics of the key continuous features.

Table 2: Descriptive statistics for key continuous features (full sample).

Feature	Mean	Std	Q1	Q3
Age (years)	38.85	11.11	29.0	46.0
Graduation year	2008.3	10.7	2002	2017
Weekly working hours	44.1	10.7	40.0	50.0
Scholarship (%)	0.58	0.49	0.0	1.0
Postgraduate degree (%)	0.57	0.49	0.0	1.0
Worked abroad (%)	0.32	0.47	0.0	1.0

3. Methodology

3.1. Experimental Setup

The dataset was divided 80/20 into stratified training sets ($n = 20,284$) and hold-out test sets ($n = 5,072$), preserving the class ratio. Model selection was performed through 5-fold stratified cross-validation in the training set. ROC-AUC was chosen as the primary metric given the class

imbalance. All models used `class_weight = 'balanced'` where available to compensate for the 93:7 ratio.

3.2. Models

Six model families were evaluated:

- **Logistic Regression** (SAGA solver, `max_iter = 500`): linear probabilistic baseline.
- **Decision Tree** (max depth 5): interpretable non-linear baseline.
- **Random Forest** (200 trees, max depth of 6, 20 min samples per leaf): ensemble via bagging.
- **Gradient Boosting** (100 estimators, max depth of 3, learning rate of 0.05, subsample of 0.8): sequential boosting ensemble.
- **Gaussian Naïve Bayes**: density-estimation classifier assuming Gaussian feature distributions per class.
- **Quadratic Discriminant Analysis (QDA)**: density-estimation with class-specific covariance matrices (`reg_param = 0.5`).

The tree depths were kept shallow to limit computational cost and reduce overfitting. Logistic Regression and density-estimation models were preceded by a `StandardScaler` in the pipeline.

3.3. Interpretability: SHAP

To go beyond the importance of features, we applied SHAP [2] using `TreeExplainer` on the winning model. SHAP decomposes each prediction into additive feature contributions, providing both global importance rankings and local, per-observation explanations.

4. Model Comparison Results

4.1. Cross-Validated Performance

Table 3 reports 5-fold cross-validated metrics in the training set. **Gradient Boosting** achieved the highest ROC-AUC (0.802 ± 0.009), followed closely by Random Forest (0.782 ± 0.008). Density-estimation approaches (QDA, Gaussian NB) performed substantially worse in terms of discriminative ability.

Table 3: 5-fold stratified cross-validation results (training set, $n = 20,284$). Models sorted by ROC-AUC.

Model	ROC-AUC	F1	Precision	Recall
Gradient Boosting	0.802 ± 0.009	0.105	0.583	0.058
Random Forest	0.782 ± 0.008	0.274	0.173	0.664
Logistic Regression	0.766 ± 0.011	0.269	0.169	0.661
Decision Tree	0.758 ± 0.010	0.251	0.155	0.661
QDA	0.659 ± 0.008	0.138	0.075	0.853
Gaussian NB	0.498 ± 0.008	0.129	0.069	0.942

The trade-off between precision and recall is pronounced

across all models: Gradient Boosting maximizes precision (0.583 CV, 0.500 hold-out) at the cost of low recall (0.058), while tree-free models recall many positives but at a high false-positive rate. For the applied use case (screening alumni for targeted outreach), high precision is preferable.

4.2. Hold-Out Test Performance

Table 4 reports the final evaluation of Gradient Boosting in the held-out test set.

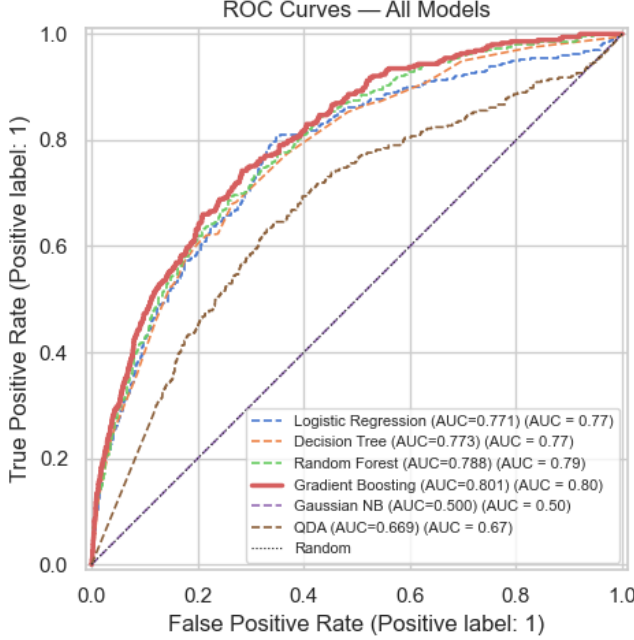


Figure 1: ROC curves for all six models evaluated on the hold-out test set ($n = 5,072$). Gradient Boosting (solid red) achieves the highest AUC of 0.80, while Gaussian NB performs at chance level ($AUC = 0.50$). The dotted diagonal represents a random classifier.

Table 4: Gradient Boosting hold-out test set performance ($n = 5,072$).

Metric	Founders	Non-founders
ROC-AUC	0.801	
Precision	0.500	0.930
Recall	0.054	0.999
F1	0.097	0.963
Support	353	4,719
Accuracy	0.930	

The ROC-AUC of 0.801 is consistent with the cross-validated estimate (0.802), indicating stable generalization without meaningful overfitting.

4.3. Statistical Significance of Individual Features

Before modeling, we conducted bivariate tests to validate the relevance of features, and Chi-square tests for binary features (Table 5) confirm that all behavioral and leadership variables are highly significant.

Table 5: Chi-square test results for binary features vs. `founded_nonprofit`.

Feature	χ^2	p -value
Volunteers	917.0	$< 10^{-200}$
Held CEO	725.9	$< 10^{-159}$
Advisory board	703.2	$< 10^{-154}$
Founded company	626.3	$< 10^{-137}$
Donates money	445.2	$< 10^{-98}$
Held VP	298.6	$< 10^{-66}$
Held director	244.7	$< 10^{-54}$
Held elected pos.	190.5	$< 10^{-42}$
Postgraduate	149.8	$< 10^{-33}$
Worked abroad	23.1	1.5×10^{-6}
Scholarship	13.6	2.3×10^{-4}

The Mann-Whitney U tests on continuous features show that age ($mean_{\text{founders}} = 46.1$ vs. 38.3 ; $p < 10^{-150}$) and years since graduation (22.4 vs. 15.2 ; $p < 10^{-131}$) are among the most significant numeric discriminators.

5. SHAP Interpretability Analysis

5.1. Global Feature Importance

Table 6 lists the top 15 features by mean absolute SHAP value in the test set. **Volunteering** alone accounts for a mean SHAP of 0.436, which is more than twice the next-highest feature, confirming it as the dominant signal. **Prior company founding** and **age** are the second and third most important predictors.

Table 6: Top 15 features by mean SHAP value (Gradient Boosting, test set $n = 5,072$).

Rank	Feature	Mean SHAP
1	Volunteers (current)	0.436
2	Founded a company	0.209
3	Age	0.174
4	Donates money	0.128
5	Held CEO role	0.097
6	Advisory board member	0.092
7	Employment: paid employee	0.074
8	Postgraduate degree	0.061
9	Sense of purpose (wb_purpose)	0.055
10	Years since graduation	0.047
11	Entrepreneurship comp. (today)	0.038
12	Entrepreneurship comp. (at grad)	0.031
13	Leadership breadth	0.028
14	School: Humanidades y Educación	0.025
15	Mother: trader occupation	0.020

5.2. Directional Effects

The summary plot of the beeswarm reveals consistent directional patterns: *volunteers*, *age*, *founded_company*, *donates_money*, *held_ceo*, and *now_comp_entrepreneurship* all have positive SHAP values for high feature values, meaning higher values of these features increase the predicted probability of founding a nonprofit. In contrast, being a paid employee (as opposed to being self-employed or running an organization) is associated with a negative SHAP contribution.

The dependence plot for *age* shows a near-monotonic positive slope: every additional decade of age corresponds to a materially larger positive SHAP contribution. The same holds for *years_since_grad* and for both entrepreneurship competency scores at graduation and today.

5.3. Individual Prediction Explanation

The highest-risk test respondent had a predicted probability of $P(\text{nonprofit}) = 97\%$. Their key feature values were: volunteer (yes), founded company (yes), age 69, donates money (yes), held CEO (yes), advisory board (yes), postgraduate degree (yes), sense of purpose 9/10, entrepreneurship competency 10/10 at both graduation and today, leadership breadth 4, and 47 years since graduation. The waterfall plot decomposition confirms that these features add cumulatively large positive SHAP values on top of the negative base value of -2.99 .

6. High-Probability Alumni Profile

We define the *high-probability* segment as alumni whose predicted score falls in the top decile (≥ 90 th percentile; threshold $P \geq 0.156$, $n = 508$). Table 7 compares their mean feature values with the remaining 90% (*rest*).

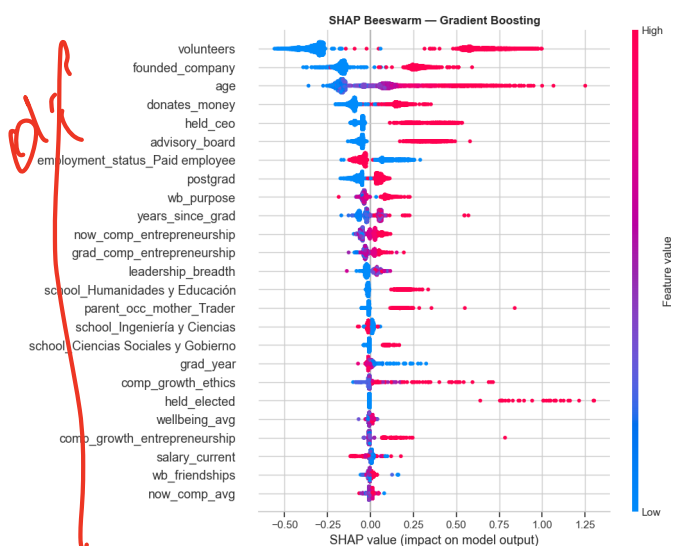


Figure 2: SHAP beeswarm plot for the Gradient Boosting model. Each point represents one test-set observation; color encodes the feature value (red = high, blue = low). Features with predominantly red points on the right side of zero increase the predicted probability of founding a nonprofit, while being a paid employee (blue, left) has a negative effect.

Table 7: Numeric profile comparison: high-probability founders (top decile) vs. rest.

Feature	High-prob.	Rest	$\Delta\%$
Age (years)	50.9	37.6	+35.5
Years since graduation	27.0	14.6	+85.1
Leadership breadth	1.64	0.52	+217.3
Postgraduate degree	0.73	0.54	+35.2
Worked abroad	0.38	0.31	+22.5
Entrepreneurship comp. (today)	8.89	7.64	+16.4
Entrepreneurship comp. (grad.)	7.70	6.68	+15.3
Sense of purpose (wb_purpose)	8.98	7.89	+13.7
Scholarship received	0.44	0.60	-27.2

Within the high-probability group, 27.8% are confirmed founders in the test set—a fourfold increase over the base rate of 6.95%.

The most striking differences are in leadership breadth (+217%), years since graduation (+85%), age (+36%), and postgraduate attainment (+35%). Interestingly, the high-probability group has a *lower* scholarship rate (−27%), suggesting that those from more resource-endowed family backgrounds are somewhat overrepresented, although the effect is weaker than the experiential predictors.

7. Discussion

7.1. Key Findings

Three thematic clusters emerge from the combined statistical and SHAP analysis:

Behavioral disposition. Volunteering and monetary donation to social causes are by far the strongest predictors (SHAP = 0.436 and 0.128, respectively). This suggests that the propensity to establish a social organization is part of a broader, long-standing civic orientation rather than a sudden decision. Alumni who volunteer are 2.1× more likely to be in the high-risk group (90.7% vs. 27.4%).

Experiential capital. Age, years since graduation, leadership breadth, and having previously founded a for-profit company are all strong positive predictors. This aligns with theories of social entrepreneurship that emphasize accumulated human, social, and financial capital [3]. The model suggests that there is no single trigger year; the probability increases gradually with professional tenure.

Competency and purpose. Entrepreneurial competence (self-assessed at graduation and today) and purpose in life (wb_purpose) are robust predictors after controlling for all demographic variables. This supports the hypothesis that the institution’s curriculum—specifically its focus on innovative entrepreneurship and ethics—leaves a measurable trace in alumni who go on to establish social organizations.

7.2. Limitations

Several limitations should be noted. First, the survey is cross-sectional and self-reported; causal inference is not possible. Second, the class imbalance (6.95% positive) limits the absolute recall of any model. Third, the volunteer and donation variables may be endogenous: alumni who founded an organization may answer the volunteering question differently. Fourth, the feature space after one-hot encoding is high-dimensional (544 features), which may introduce noise despite regularization. Finally, SHAP attributions indicate feature importance within the model, not necessarily true causal pathways.

8. Conclusion

We developed an end-to-end machine learning pipeline to predict the founding of social organizations among Tecnológico de Monterrey alumni. Gradient Boosting achieved the best performance given (ROC-AUC = 0.801), representing a 30 percent improvement over random guessing. The SHAP analysis identifies volunteering, prior entrepreneurial experience, age, donation habits, and leadership history as the dominant predictors. Academic factors (postgraduate education, self-assessed entrepreneurial competence, and sense of purpose in life) contribute independently above and beyond the career trajectory.

For institutional practice, these findings suggest that alumni most likely to create social organizations can be identified during their studies by monitoring three observable proxy signals: (1) participation in volunteer programs, (2) high scores on entrepreneurship competency

assessments, and (3) a reported sense of personal purpose. These signals could inform targeted fellowship programs, social-impact accelerators, and alumni engagement initiatives.

Future work should explore longitudinal panel data, causal modeling, and text-based features (open-ended survey responses) to further improve prediction and deepen the understanding of pathways to social entrepreneurship.

References

- [1] Quacquarelli Symonds (QS). *Global Impact of Tecnológico de Monterrey Alumni: 80th Anniversary Study*. QS Intelligence Unit, 2023.
- [2] Lundberg, S. M., & Lee, S.-I. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, 30, 2017.
- [3] Dacin, M. T., Dacin, P. A., & Tracey, P. Social entrepreneurship: A critique and future directions. *Organization Science*, 22(5), 1203–1213, 2010.
- [4] Austin, J., Stevenson, H., & Wei-Skillern, J. Social and commercial entrepreneurship: Same, different, or both? *Entrepreneurship Theory and Practice*, 30(1), 1–22, 2006.
- [5] Breiman, L. Random forests. *Machine Learning*, 45(1), 5–32, 2001.
- [6] Friedman, J. H. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232, 2001.

B.4.0 Inference

no far an
biscar may recent